

Synoptic CB: Porewater DOC, TN

June 2022 Samples

2026-06-17

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##Setup - Change things here & write any notes

```
#identify section
cat("Setup Information")
```

Setup Information

```
##### Run information - PLEASE CHANGE
Date_Run = "06/15/22" #Date that instrument was run
Run_by = "Stephanie J. Wilson" #Instrument user
Script_run_by = "Stephanie J. Wilson" #Code user
run_notes = "Standard Curve assessed manually on the instrument." #any notes from the run
samples <- c("GCW", "GWI", "MSM", "SWH") #whatever identifies your samples within the same names

samples_pattern <- paste(samples, collapse = "|")
#samples_pattern <- "GCW" #use this instead of the line above if you have only one site code
chks_name = "CKSTD" #what did you name your check standards?

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name = c("Raw Data/TOCTN_COMPASS_Synoptic_TCTN_202206_1.txt" , "Raw Data/TOCTN_COMPASS_Synoptic_TCTN_202206_1.txt")
```

```

#file path and name for raw all peaks file
# raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202505_allpeaks.txt"

#file path and name of processed data file
processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DOC_202206.csv"

##### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2022.csv"

#qac log file path for this year
#Log_path = "Raw Data/COMPASS_Synoptic_TOCTN_QAClog_2022.csv"

```

```
##Set Up Code
```

```
##Read in metadata and create similar sample IDs for matching to samples
```

0.1 Import Data Functions

0.2 Import Sample Data

```
## Import Sample Data
```

```
## New names:
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## # A tibble: 6 x 4
```

sample_name	npoc_raw	tdn_raw	run_datetime
<chr>	<dbl>	<dbl>	<chr>
1 MSM_WC_PW_MonMon_DOC_20220607_SipA_20cm	21.0	0.870	6/15/2022 2:20:43 AM
2 MSM_WC_PW_MonMon_DOC_20220607_SipB_20cm	9.43	0.506	6/15/2022 2:47:44 AM
3 MSM_WC_PW_MonMon_DOC_20220607_SipC_10cm	15.1	0.826	6/15/2022 3:14:26 AM
4 MSM_WC_PW_MonMon_DOC_20220607_SipB_10cm	8.35	0.484	6/15/2022 3:39:18 AM
5 MSM_WC_PW_MonMon_DOC_20220607_SipA_10cm	21.2	0.931	6/15/2022 4:07:46 AM
6 MSM_WC_PW_MonMon_DOC_20220607_SipC_20cm	18.5	1.57	6/15/2022 4:34:40 AM

```
##Fix sample naming
```

```

#metadata: 202207_GWI_UP_LysB_20cm
#samples: MSM_WC_PW_MonMon_DOC_20220607_SipA_20cm
dat_raw <- dat_raw %>%
  mutate(sample_name = gsub("MSM_UP_PW_MonMon_DOC_20220607", "20220607_MSM_UP", sample_name)) %>%
  mutate(sample_name = gsub("MSM_WC_PW_MonMon_DOC_20220607", "20220607_MSM_WC", sample_name))

##just need the sample year and month
dat_raw <- dat_raw %>%
  mutate(sample_name = gsub("20220607", "202206", sample_name))

```

0.3 Assessing Standard Curves - assessed manually on the instrument

0.4 Assess Check Standards

```
## Assess the Check Standards
```

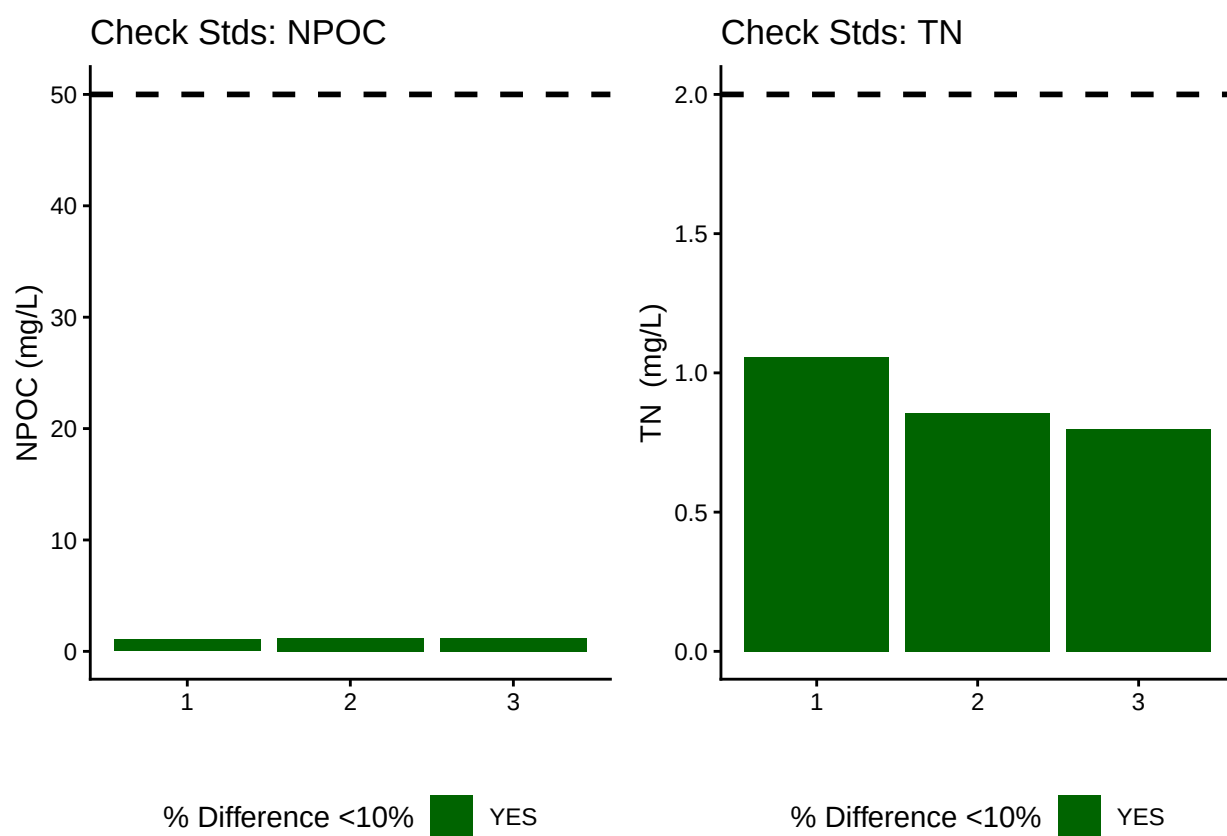
```
## New names:
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## [1] "Carbon Check Standard RSD within Range"
```

```
## [1] "Nitrogen CHECK STANDARD RSD TOO HIGH - REASSESS"
```



```
## [1] ">60% of Carbon Check Standards are within range of expected concentration"
```

```
## [1] ">60% of Nitrogen Check Standards are within range of expected concentration"
```

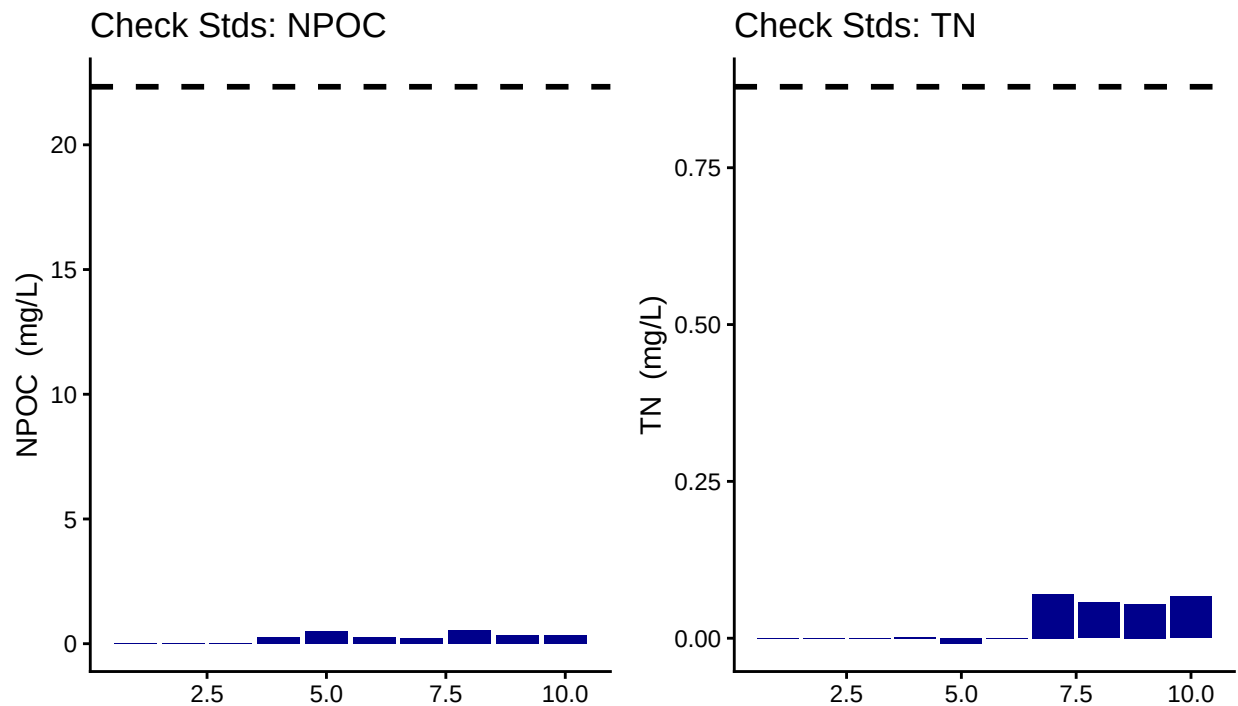
0.5 Assess Blanks

```
## Assess Blanks
```

```
## New names:
## New names:
## * ' ' -> '...14'
```

```
## [1] ">60% of Carbon Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">60% of Nitrogen Blank concentrations are lower 25% quartile of samples"
```



Blank Conc <25% Quartile Samples ■ YE

Blank Conc <25% Quartile Samples ■ Y

```
## carbon blanks:
```

```
## [1] 0.25147
```

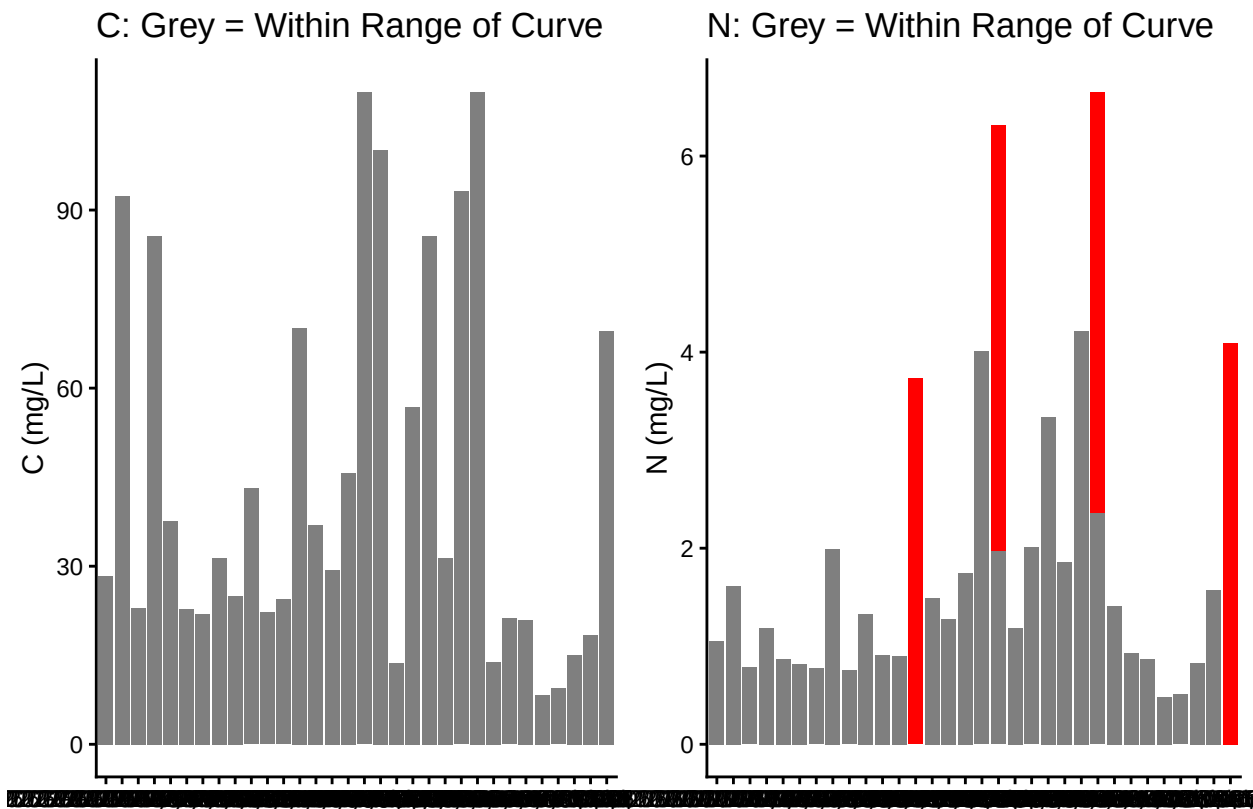
```
## nitrogen blanks:
```

```
## [1] 0.023999
```

0.6 Assess Duplicates - no duplicates ran

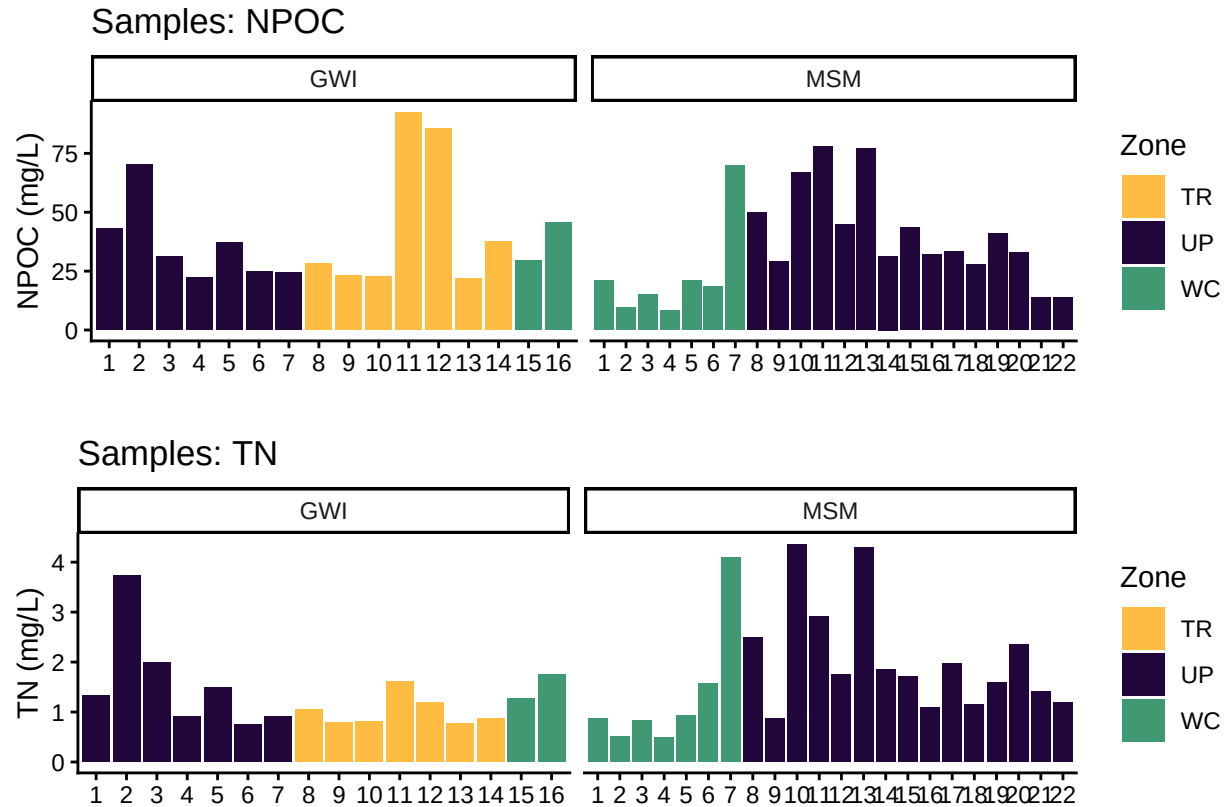
0.7 Sample Flagging - Are samples Within the range of the curve?

```
## Sample Flagging
```



0.8 Visualize Data by Plot

```
## Visualize Data
```



0.9 Convert data from mg/L to uMoles/L

0.10 Check to see if samples run match metadata & merge info

```
## Check Sample IDs with Metadata

## Some sample IDs are missing from metadata.

## [1] "202206_MSM_WC_SipA_20cm" "202206_MSM_WC_SipB_20cm"
## [3] "202206_MSM_WC_SipC_10cm" "202206_MSM_WC_SipB_10cm"
## [5] "202206_MSM_WC_SipA_10cm" "202206_MSM_WC_SipC_20cm"
## [7] "202206_MSM_WC_SipC_45cm" "202206_MSM_UP_LysB_45cm"
## [9] "202206_MSM_UP_LysC_45cm" "202206_GWI_TR_LysB_45cm"
## [11] "202206_GWI_WC_SipA_20cm" "202206_GWI_WC_SipC_20cm"
```

0.11 Export Processed Data

```
## Export Processed Data

## # A tibble: 6 x 21
##   Project      Region Site Zone Replicate Depth_cm Sample_ID Year Month Day
##   <chr>        <chr> <chr> <chr> <chr>      <chr>      <chr>    <int> <int> <int>
## 1 COMPASS: Sy~ CB     MSM   WC     SipA      20      202206_M~    NA    NA    NA
```

```
## 2 COMPASS: Sy~ CB      MSM   WC      SipB      20      202206_M~      NA      NA      NA
## 3 COMPASS: Sy~ CB      MSM   WC      SipC      10      202206_M~      NA      NA      NA
## 4 COMPASS: Sy~ CB      MSM   WC      SipB      10      202206_M~      NA      NA      NA
## 5 COMPASS: Sy~ CB      MSM   WC      SipA      10      202206_M~      NA      NA      NA
## 6 COMPASS: Sy~ CB      MSM   WC      SipC      20      202206_M~      NA      NA      NA
## # i 11 more variables: Time <lgl>, Time_Zone <lgl>, npoc_mgL <dbl>,
## #   npoc_uM <dbl>, npoc_flag <chr>, tdn_mgL <dbl>, tdn_uM <dbl>,
## #   tdn_flag <chr>, Analysis_runtime <chr>, Run_notes <chr>, Field_notes <chr>
```

```
#end
```